

Study Questions: Sharing the Vision

Analytical Questions

1. Active Inference frames communication as generative model alignment between agents. Explain what this means in concrete terms for an inventor pitching their product to an investor who has no background in the invention's domain. What must the inventor infer about the investor's current model? What model updates must the pitch achieve?
2. Why is leading a pitch with technical details often ineffective? Use Active Inference concepts (prediction error, model alignment, prior expectations) to explain why the problem-first structure works better for non-technical audiences.
3. Compare the communication strategies needed for three different audiences of the same medical device invention: (a) a regulatory body, (b) a surgeon who will use it, and (c) a patient on whom it will be used. How does the externalized model differ for each audience? What causal detail is essential for each, and what can be omitted?
4. Steve Jobs's iPhone launch demonstration is an example of communication through shared sensory evidence. Analyze why demonstration was more effective than description for this particular invention. Under what conditions is demonstration superior to verbal description, and when is the reverse true?
5. IKEA's wordless assembly instructions build the user's generative model incrementally. Explain how each instruction step functions as a "managed prediction error" — providing enough novelty to update the model without overwhelming the user. What happens when instruction steps are too large (too much model update at once) or too small (insufficient new information)?
6. Jonas Salk's communication strategy for the polio vaccine addressed factual, causal, emotional, and trust dimensions. Why was each dimension necessary? Could Salk have succeeded by communicating only the scientific facts, without addressing emotional and trust dimensions?

7. The concept of affordances — design features that communicate function through form — represents communication embedded in the artifact itself. Analyze three everyday objects and identify how their affordances communicate proper use. Then identify one case where poor affordance design leads to user error.
8. Patent applications follow a specific communication structure: prior art, problem statement, invention description, claims. Analyze this structure as a model alignment strategy. Who is the intended reader? What is their assumed prior model? How does each section update their model?
9. In multi-agent invention teams (Module 2), communication serves to align team members' generative models. What happens when team members have different precision weightings on the same evidence — for example, an engineer who trusts test data more than user feedback, and a designer who trusts user feedback more than test data? How can communication resolve this precision conflict?
10. Explain how user instruction design relates to the Active Inference concept of building a "good enough" model — one that enables effective action without requiring full understanding of the underlying mechanism. When is a simplified user model better than a complete one? Can an overly complete model actually harm user performance?

Applied Questions

1. Write a 60-second pitch for your invention to a non-technical audience. After writing it, identify: (a) the prediction error you create (the problem statement), (b) the model update you provide (the solution), (c) the preference alignment (the value proposition), and (d) the credibility signal.
2. Now rewrite the same pitch for a technical expert in your domain. What changes? What stays the same? What technical vocabulary can you add? What framing do you change?
3. Design the first page of a user manual for your invention using the "incremental model building" approach. Each instruction should add exactly one new piece of information to the user's model. Test this on someone unfamiliar with your invention — can they follow it?

4. Identify three affordances in your invention design that communicate function without words. If your invention has no affordances yet, propose three design changes that would create them.
5. Think about a time when someone tried to explain something complex to you and failed. What went wrong from an Active Inference perspective? Was the model gap too large? Was the precision weighting wrong? Was the sequencing poor? How would you restructure the explanation?
6. Observe a product demonstration (live, video, or in a store). Analyze it as a communication event: what prediction errors does the demonstrator create in the audience? What model updates do they achieve? Where does the demonstration succeed and fail?
7. If you had to describe your invention in only one image (no words), what image would you choose? Why? What does this exercise reveal about the core of your invention's value proposition?
8. Consider the feedback loop in communication: the audience's response is evidence about whether your message succeeded. Design a communication experiment where you deliver the same pitch to two different people, observe their responses, and update your pitch model based on the prediction errors. What specific audience reactions will you look for?
9. Your invention is being misused by users — they are using it in a way you did not intend, leading to poor results. Using Active Inference, diagnose the communication failure: is the user's generative model of the invention wrong? If so, where is the specific model misalignment? How would you fix it through design changes, instruction changes, or both?
10. Write a one-paragraph "model alignment statement" for your invention that could serve as the foundation for all communication — pitches, documentation, instructions, and marketing. This statement should capture: the problem (prediction error), the mechanism (causal model), the value (preference alignment), and the user's role (action model).